Fiscal Year:	FY 2023	**Task Last Updated:**	FY 04/03/2023
PI Name:	Eslami, Ali Ph.D.		
Project Title:	Probabilistic Approach to Reverse Engineering of Spaceflight Molecular Networks		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	67260	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	EPSCoR-Space Biology NNH20ZHA001C
Start Date:	07/01/2020	**End Date:**	01/31/2022
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA EPSCoR
Contact Monitor:	Compton, Jeppie	**Contact Phone:**	
Contact Email:	jeppie.r.compton@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 1/31/2022 (original end date was 6/30/2021) per NSSC information (Ed., 1/18/22)		
Key Personnel Changes/Previous PI:	Dr. Leonard Scott Miller serves as the Kansas NASA EPSCoR Director, he/she is the administrative Principal Investigator on this proposal and will oversee the overall progress of the work and report to NASA.		
COI Name (Institution):	Broddick, Jared Ph.D. (NASA Ames Research Center)		
Grant/Contract No.:	80NSSC20M0133		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Established Program to Stimulate Competitive Research (EPSCoR) Project Reverse engineering of molecular networks is one of the most challenging tasks in systems biology and bioinformatics. This is mainly due to the complex nature of biological systems that involve many factors and uncertainties. However, with the rapid biotechnological advancements, large-scale high-throughput biological data have become available. These data have enabled researchers to deduce and understand how interactions among the vast array of components in biological systems relate and affect each other. It is a general consensus among system biologists and bioinformaticians that such interactions form networks that aim at capturing the dependencies between the interacting entities. Therefore, existing literature provides a strong prior biological knowledge to conduct research in this area. However, for Space Biology there is very little research focus on the development of computational and modeling tools that enable the discovery and inference of space environment-specific molecular networks. Although the ground-based system biology computational tools can be adapted for reverse engineering spaceflight molecular networks, they are largely inadequate to capture biophysical parameters relating to space environment. Thus, the goal of this research is to develop an approach for reconstructing, inferring, and analyzing spaceflight molecular networks using data from the NASA GeneLab data system. In addition, we adapt our preliminary probabilistic graphical model tool to validate and quantify how well the inferred networks explain the observed experimental data. Intellectual merit: The proposed work seeks to grow a research area crossing frontiers in probabilistic methods, graph theory, algorithms, biology, and data science. Focusing on the reconstruction, inference, and analysis of spaceflight molecular networks for which very little literature exists, the work has the potential to advance knowledge and understanding across several related fields. Furthermore, it provides an opportunity to translate decades of biological data from experiments performed in space environments into information that fosters innovation in Space Biology. Specifically, the significance of the proposed activity is as follows: • Developing computational tools for the discovery of biological networks influenced by space conditions. • Establishing mathematical framework for the analysis and validation of spaceflight biological networks. • Developing algorithmic tools for generating biologically realistic spaceflight expression data. • Predicting biochemical and kinetic parameters of biological processes in spaceflight. Broader impacts: The Principal Investigator believes that the proposed work has far-reaching implications on basic science, education, and technology that will benefit society, in general. The broader impacts include 1) Advancing understanding of key biological mechanisms that enable biological organisms to thrive in space environments, 2) Training of graduate students in bioinformatic data analysis, and 3) Integration of research results in the classes taught by the Principal Investigator and to improve the curriculum at Wichita State University.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	• We introduced a computational framework that combines the formulation of Boolean networks (BNs) and factor graphs to explore the global dynamical features of biological systems. A message-passing algorithm was proposed for this formalism to evolve network states as messages in the graph. The model was applied to assess the network state progression and the impact of gene deletion in the budding yeast cell cycle. Simulation results show that our model predictions match published experimental data. • We investigated the prediction of transcription profiles of genes exposed to conditions in space from their nucleotide sequence. To do this, we collected differentially and non-differentially expressed genes (DEGs/NDEGs) of Mus-based experiments on the NASA GeneLab database. Then, we engineered nucleotide-based features capturing gene length, number of exons, GC-content, interaction with known transcription factors, state of chromatin, annotated enhancer characteristics, interactions with miRNAs, and RNA-binding protein status. Subsequently, we used the 30% top and least differentially expressed genes, resulting in a total of 3,272 genes with an equal number of DEGs and NDEGs, to train a deep neural network to predict if a gene is differentially expressed or not given the engineered features. Both models trained with all features and a selection of the most relevant features achieved true positive rates above 0.7 for all false positive rates, with AUCs of 0.927 and 0.934, respectively. Similarly, a minimum of 0.65 precision was achieved for all recall values, with most recall values achieving perfect precision. The result demonstrates that the most differentially expressed genes can be predicted from genomically encoded features. • We developed a time-series approach to bacteria gene mutation prediction, where mutation occurrence is modeled sequentially. In our approach, the most connected co-functional genes are predicted first, and used as additional input features in subsequent predictions. We trained a neural network to predict gene mutations in Escherichia coli from the exposure conditions. Results on out-of-sample tests on adaptive laboratory evolution experiments curated from the literature showed that the co-functional network-based features improved the predictive performance by a maximum 33.53% and 45.18% in the area under the receiver operating characteristic curve (AUC) and area under the precision-recall curve (AUPRC), respectively. Our approach is beneficial in the development of drugs and vaccines.
Bibliography Type:	Description: (Last Updated: 04/03/2023)
Articles in Peer-reviewed Journals	Kotiang S, Eslami A. "Density evolution for noise propagation analysis in biological networks." IEEE Access. 2022 Jan;10:4261-70. http://dx.doi.org/10.1109/ACCESS.2022.3140720 , Jan-2022
Articles in Peer-reviewed Journals	Kotiang S, Eslami A. "Boolean factor graph model for biological systems: the yeast cell-cycle." BMC Bioinformatics. 2021 Sep 17;22:442. http://dx.doi.org/10.1186/s12859-021-04361-8 , Sep-2021
Articles in Peer-reviewed Journals	Okwori M, Eslami A. "Investigating the impact of gene cofunctionality in predicting gene mutations of E. coli." IEEE Access. 2020 Sep 14;8:167397-410. http://dx.doi.org/10.1109/ACCESS.2020.3023662 , Sep-2020